

How the Dumbarton forward ratings work

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Every player gets two ratings out of 100. **Right now** measures how productive he has been over recent seasons, once league strength is accounted for. **Going forward** takes that same number and adjusts it for age and form, to suit a signing you want for several seasons. Both are calculated by fixed rules from public statistics. Nobody types a rating in by hand.

The four steps

1. **Collect league seasons.** Goals, assists and games per player per season come from FotMob. Only league games count, and only seasons with at least 10 games.
2. **Put every league on one scale.** Each league gets an exchange rate against League Two, learned from players who moved between leagues. A player's goals and assists per game are multiplied by that rate.
3. **Summarise his recent career.** Up to four counted seasons are blended into one "output level", with recent seasons and seasons with more games counting for more. His best season, form and any recent step up are worked out alongside.
4. **Turn that into ratings.** His output level is compared with League Two forwards and combined with games played to give the Right now rating. Age and form adjust that into the Going forward rating.

Contract status from Transfermarkt sits alongside the ratings but never changes them.

Step 1: What data goes in

A season only counts towards a rating if it is a league season with at least 10 games, in a league that has an exchange rate.

- **Source.** FotMob player pages give, for each season and competition: club, games, goals and assists. Minutes are not available for every season at this level, so everything is measured **per game**, not per 90 minutes.
- **Output measure.** Goals plus assists, divided by games. A striker with 12 goals and 6 assists in 30 games scores 0.60.

- **League games only.** Any competition with "Cup" in its name, or flagged as a cup, is left out. Where a player appears in two league competitions in one season, the one with more games is used.
- **10-game minimum.** Seasons with fewer than 10 games are kept in his record and shown on his profile, but do not count towards his rating. This stops a hot first month from dominating.
- **Who is assessed.** Players whose FotMob position mentions forward, striker, attacking or winger.
- **Identity is always current.** Club, league and age shown on the list always come from his most recent season, even when that season doesn't count yet.

Step 2: League exchange rates

Each league's goals and assists are converted to a League Two equivalent by multiplying by that league's rate. League Two is fixed at 1.00.

$$\text{adjusted per game} = \text{actual per game} \times \text{league rate}$$

League	Rate vs League Two	Players moved (evidence)	Effect
Highland/Lowland League	0.67	16	Output scaled down by a third
League Two	1.00	baseline	None
League One	1.32	94	Output scaled up by about a third
Championship	1.38	34	Output scaled up by just over a third
Premiership	2.52	7	Output more than doubled

Figures are from the run of 30 September 2026 and are recalculated on every rebuild.

How the rates are learned. The rates come only from players who changed league between two consecutive seasons. To count as evidence, a move must:

- fall within the last six seasons;

- have at least 5 league games on both sides of the move;
- show some goal or assist output on both sides.

Every qualifying move says something like "this player produced 0.45 per game in the Highland League and 0.30 in League Two the next season". All moves are solved together for the set of rates that best fits them. Moves with more games on both sides carry more weight.

When a league gets no rate. A league needs at least 5 qualifying moves before its rate is published. Leagues below that, such as the Irish Premier Division or the USL, show on a player's record but don't count towards his rating. The Premiership rate rests on only 7 moves, so treat it as rough.

Step 3: Summarising his recent career

A player's **output level** is a weighted average of his adjusted goals and assists per game, across his last four counted seasons.

$$\text{output level} = \frac{\sum_s w_s \times \text{adjusted per game}_s}{\sum_s w_s}, \quad w_s = 0.55^{\text{years back}} \times \text{games}_s$$

In words: each season's weight is its games played, multiplied by 0.55 for every year it sits before his most recent counted season.

Season	Recency multiplier	Weight of a 30-game season
Most recent counted	1.00	30.0
One year earlier	0.55	16.5
Two years earlier	0.30	9.1
Three years earlier	0.17	5.0

So last season counts roughly twice as much as the one before, and a 10-game season counts a third as much as a 30-game one. The same method gives the "before adjustment" figure, using his actual numbers.

Best season. His highest adjusted goals and assists per game in any counted season. Because both use the same 10-game bar, his best season can never sit below his output level.

Form. His most recent season of 12 or more games is compared with the weighted average of his earlier 12-plus-game seasons.

- 15% or more above: **Improving**
- 15% or more below: **Dropping off**
- In between: **Steady**
- Fewer than two such seasons: **Too early to tell**

Stepped up. Flagged when the league he plays in now has a rate at least 5% higher than the league of his most recent counted season elsewhere, provided he played at least 8 games there. It works from his current league even when he hasn't yet played 10 games there.

Cautions shown on his profile.

- *Not many games to go on:* fewer than 20 games across his counted seasons.
- *Best form came in a weaker league:* his best season was in a league with a rate of 0.70 or below.

Step 4: The two ratings

The **Right now** rating is 70% output and 30% games played, both measured against forwards currently in League Two. **Going forward** multiplies that by an age factor and a form factor.

Output score (0 to 1). The share of current League Two forwards whose output level is at or below his. If he out-produces 80% of them, his output score is 0.80. Anyone at or above the best of them scores 1.00.

Games score (0 to 1). His games across counted seasons, divided by the typical (median) figure for League Two forwards, capped at 1. Matching or beating the typical figure scores 1.00.

$$\text{Right now} = 100 \times \min\big(1, 0.7 \times \text{output score} + 0.3 \times \text{games score}\big)$$

$$\text{Going forward} = 100 \times \min\big(1, \text{Right now base} \times \text{age factor} \times \text{form factor}\big)$$

Both are rounded to the nearest whole number. Several players can show 100 when they out-produce every League Two forward and have a full record of games.

Age factor. A rough guide to how many years of peak value a forward has left.

Age	Age factor
19 or under	0.90
20 to 26	1.00
27 to 28	0.92
29 to 30	0.82
31 to 32	0.70
33 to 34	0.58
35 or over	0.48
Age unknown	0.90

Under-20s are slightly discounted because their output is less proven, not because they lack upside.

Form factor.

Form	Form factor
Improving	1.08
Steady	1.00
Too early to tell	1.00
Dropping off	0.90
Stepped up (overrides the above if higher)	1.15

The weights are fixed in the shared site so everyone sees the same ranking. Changing them is a one-line edit if the club wants a different balance.

Contract status

Contract status is shown next to the ratings but has no effect on them. It comes from Transfermarkt squad pages, currently for League Two clubs only.

Matching players across the two sites. FotMob and Transfermarkt don't share player IDs, so each Transfermarkt player is matched to a FotMob player by name and date of birth. Names are compared with accents, capitals and punctuation stripped.

1. Same name and exact same date of birth: matched.
2. If FotMob has no date of birth: same name and birth year within one year, only if exactly one player fits.
3. If there's only one player of that name and no date to check: matched on name alone.
4. Two or more possible players that can't be told apart: left unmatched rather than guessed.

In the 30 September run, 57 of 109 Transfermarkt players matched, all on exact date of birth. The rest show as "Not found".

Shown as	Rule
Contract ended	End date has already passed. He may be a free agent, or the record may be out of date.
Ending soon	Ends within the next 8 months
Under contract	Ends more than 8 months away
Date unclear	Matched, but Transfermarkt lists no end date
Not found	No confident Transfermarkt match

The status is worked out when `match_market.py` runs, not live. Rerun it with each data refresh so "Ending soon" stays accurate as time passes.

Worked example: Lennon Walker

Walker's real record gives a Right now rating of about 79 and a Going forward rating of about 91. Two inputs below are marked as illustrative, because they depend on the whole League Two pool on the day; the site shows his exact figures.

His record. 2025/26 at Caledonian Braves in the Highland/Lowland League: 32 games, 27 goals, 4 assists. 2026/27 so far at Edinburgh City in League Two: 8 games, 3 goals, 2 assists. He is 22.

1. **Counted seasons.** Only 2025/26 counts. 2026/27 has 8 games, under the 10-game minimum, so it shows as "this season so far" instead.

2. **Actual per game.** $(27 + 4) \div 32 = 0.97$.
3. **Adjusted per game.** 0.97×0.67 Highland/Lowland rate = 0.65. With one counted season, this is also his output level and his best season.
4. **Form and step up.** Too early to tell, with only one counted season. He has stepped up: League Two's rate of 1.00 is well over 5% higher than 0.67.
5. **Output score** (illustrative). Suppose 0.65 is at or above 90% of current League Two forwards: 0.90.
6. **Games score** (illustrative). Suppose the typical League Two forward has 60 counted games: $32 \div 60 = 0.53$.
7. **Right now.** $0.7 \times 0.90 + 0.3 \times 0.53 = 0.79$, shown as **79**.
8. **Going forward.** 0.79×1.00 (age 22) $\times 1.15$ (stepped up) = 0.91, shown as **91**.

His profile also shows the caution "Best form came in a weaker league", because the Highland/Lowland rate is below 0.70.

For comparison, a steady 34-year-old with the same 0.79 base would score $0.79 \times 0.58 \times 1.00 = 0.46$, so **46** going forward. Same output, very different signing.

Limits

The ratings filter a wide field down to names worth watching. They cannot rank closely matched players with confidence.

- **Goals and assists only.** Two strikers with the same tally can play very differently. Separating them needs paid match data such as Wyscout.
- **Small samples.** A League Two season is 36 games, so a difference of 3 or 4 goals is partly luck. Read 88 and 84 as "both strong".
- **League rates are averages.** They are learned from players good enough to earn a move, which tends to flatter weaker leagues slightly. Rates with fewer moves behind them (Premiership: 7) are rougher.
- **Team context is invisible.** A forward in a dominant side gets more chances than one in a struggling side.
- **The age and form factors are judgement calls.** They follow the general pattern of forwards peaking in their mid-twenties, but the exact values are set by us, not learned from data.
- **Contract dates can be stale.** Always confirm before approaching a player.

Every setting in one place

Setting	Value	Where it lives
Games for a season to count	10	<code>store_pipeline.py</code>
Games for a season to set form	12	<code>store_pipeline.py</code>
Seasons blended into output level	4 most recent counted	<code>store_pipeline.py</code>
Recency multiplier per year back	0.55	<code>store_pipeline.py</code>
Form threshold	±15% vs earlier seasons	<code>store_pipeline.py</code>
Stepped-up threshold	Current league rate 5% higher, earlier league 8+ games	<code>store_pipeline.py</code>
"Not many games" caution	Under 20 counted games	<code>store_pipeline.py</code>
"Weaker league" caution	Best season in a league rated 0.70 or below	<code>store_pipeline.py</code>
Games per side for a league move to count	5	<code>equivalency_real.py</code>
Window for league moves	Last 6 seasons	<code>equivalency_real.py</code>
Moves needed to publish a league rate	5	<code>store_pipeline.py</code>
Output vs games weighting	70% / 30%	<code>site_template.html</code>
Age factors	See Step 4 table	<code>site_template.html</code>
Form factors	See Step 4 table	<code>site_template.html</code>
"Ending soon" contract window	8 months	<code>match_market.py</code>